25900355 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCECT202 - INTERNET OF THINGS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. What is the primary function of the power source in an IoT system? [K] / [CO1]

   a). To collect data from the environment
   b). To process data
   c). To provide power to the device
   d). To display information to the user

2. What type of wireless connectivity does the ESP32 NodeMCU support? [U] / [CO1]

   a). Only Wi-Fi
   b). Only Bluetooth
   c). Both Wi-Fi and Bluetooth
   d). Neither Wi-Fi nor Bluetooth

3. What should you do if Arduino IDE shows "COM port not available"? [U] / [CO1]

   a). Install or reinstall USB-to-Serial drivers
   b). Check USB cable connection
   c). Try a different USB port
   d). All of the above

4. Which of the following is an output device? [K] / [CO2]

a). Buzzer b). DHT11

c). LDR d). PIR sensor

5. DHT11 sensor is used to measure: [K] / [CO2]

a). Light intensity b). Motion

c). Temperature & Humidity d). Distance

6. MQ-series sensors are used for detecting: [K] / [CO2]

a). Gas leakage b). Temperature

c). Rainfall d). Motion

7. What is the Arduino command to initialize serial communication on ESP32? [A] / [CO2]

a). Serial.connect() b). Serial.start(baudrate)

c). Serial.begin(baudrate) d). Serial.init(baudrate)

8. What is the typical range of BLE in ESP32 under normal conditions? [K] / [CO3]

a). 1-10 meters b). 10-50 meters

c). 100-200 meters d). Above 1 km

9. Which ESP32 function is used to define a route for reporting? [K] / [CO3]

a). server.on() b). server.begin()

c). server.send() d). WiFi.begin()

10. Which HTTP method is used to request data from a server? [K] / [CO3]

a). POST b). PUT

c). GET d). DELETE

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. Explain IOT concept, highlighting its key components and providing examples of its applications. [U] / [CO1]

12. Mention some common Arduino APIs used for programming ESP32 and their functions. [U] / [CO1]

13. What is the use of Serial.print() function in ESP32? Explain with a simple example [K] / [CO2]

14. Write the syntax of analogRead() function and explain its use [K] / [CO2]

15. Discuss the advantages and limitations of using gas sensors in indoor air quality monitoring. [E] / [CO2]

16. List the advantages of connecting ESP32 to a local web server. [U] / [CO3]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Discuss the relevance of internet of things (IOT) in modern society, including its benefits and challenges. [U] / [CO1]

18. Apply the concept of servo motors and DC gear motors to design a simple robotic arm mechanism. [A] / [CO2]

19. Write a detailed Arduino program for ESP32 to read a push-button state and turn on an LED only when the button is pressed. Include explanations of key functions used. [A] / [CO2]

20. Describe the advantages and limitations of ESP32 Wi-Fi networking compared to other IoT connectivity options. [U] / [CO3]

[10x2 = 20]